

Camera Calibration

魏家博 (Chia-Po Wei)

Department of Electrical Engineering

National Sun Yat-sen University

Definition of Camera Calibration

- Given $\mathbf{m}_i, \mathbf{M}_i$ for $i = 1, 2, \dots, n$
 - $\mathbf{m}_i$ is the i th detected 2D landmark of shape (2,1)
 - $\mathbf{M}_i$ is the i th 3D landmark of shape (3,1)
- Find $\mathbf{K}$ and $(\mathbf{R}, \mathbf{t})$ such that
 - $\tilde{\mathbf{m}}_i = \mathbf{K}[\mathbf{R} \ \mathbf{t}]\tilde{\mathbf{M}}_i$ with $\tilde{\mathbf{M}}_i = \begin{bmatrix} \mathbf{M}_i \\ 1 \end{bmatrix}, \tilde{\mathbf{m}}_i = \lambda_i \begin{bmatrix} \mathbf{m}_i \\ 1 \end{bmatrix}$ for $i = 1, 2, \dots, n$
 - $\mathbf{R}$ is a rotation matrix, i.e. $\mathbf{R}^T \mathbf{R} = \mathbf{I}$.
 - $\mathbf{R}$ is of shape (3,3), $\mathbf{t}$ is of shape (3,1)
 - $\mathbf{K}$ is the intrinsic matrix of shape (3,3)
 - Intrinsic parameters: $\alpha_x, \alpha_y, s, x_0, y_0$

Camera Calibration with Lens Distortion

- Given $\mathbf{m}_i = [x_i, y_i]^T$, $\mathbf{M}_i = [X_i, Y_i, Z_i]^T$ for $i = 1, 2, \dots, n$
- Find $\mathbf{K}$, $(\mathbf{R}, \mathbf{t})$, (k_1, k_2, p_1, p_2) such that
 - $\tilde{\mathbf{m}}_i = L(\mathbf{K}[\mathbf{R} \ \mathbf{t}]\tilde{\mathbf{M}}_i)$ with $\tilde{\mathbf{M}}_i = \begin{bmatrix} \mathbf{M}_i \\ 1 \end{bmatrix}$, $\tilde{\mathbf{m}}_i = \lambda_i \begin{bmatrix} \mathbf{m}_i \\ 1 \end{bmatrix}$ for $i = 1, 2, \dots, n$
- The nonlinear function L models lens distortion as below
 - Radial distortion

$$x_{distorted} = x(1 + k_1r^2 + k_2r^4)$$

$$y_{distorted} = y(1 + k_1r^2 + k_2r^4)$$

- Tangential Distortion

$$x_{distorted} = x + (2p_1xy + p_2(r^2 + 2x^2))$$

$$y_{distorted} = y + (p_1(r^2 + 2y^2) + 2p_2xy)$$

Remarks

- Find $\mathbf{K}$ and $(\mathbf{R}, \mathbf{t})$ such that
 - $\tilde{\mathbf{m}}_i = \mathbf{K}[\mathbf{R} \ \mathbf{t}]\tilde{\mathbf{M}}_i$ with $\tilde{\mathbf{M}}_i = \begin{bmatrix} \mathbf{M}_i \\ 1 \end{bmatrix}$, $\tilde{\mathbf{m}}_i = \lambda_i \begin{bmatrix} \mathbf{m}_i \\ 1 \end{bmatrix}$ for $i = 1, 2, \dots, n$
 - $\mathbf{R}$ is a rotation matrix, i.e. $\mathbf{R}^T \mathbf{R} = \mathbf{I}$.
 - $\mathbf{R}$ is of shape (3,3), $\mathbf{t}$ is of shape (3,1)
 - $\mathbf{K}$ is the intrinsic matrix of shape (3,3)
- Questions
 - Is this problem linear or nonlinear?
 - How many variables are there to be solved?

Geometric Camera Calibration

Given $\mathbf{K}$, (x_i, y_i) , (X_i, Y_i, Z_i) for $i = 1, 2, \dots, n$.

Solve $(\mathbf{R}, \mathbf{t})$ such that

$$\tilde{\mathbf{m}}_i = \mathbf{K}[\mathbf{R} \ \mathbf{t}]\tilde{\mathbf{M}}_i$$

Define $\mathbf{P} = \mathbf{K}[\mathbf{R} \ \mathbf{t}]$, then

$$\begin{bmatrix} \lambda_i x_i \\ \lambda_i y_i \\ \lambda_i \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix}$$

Solve the above using SVD to obtain $\mathbf{P}$.

Perform QR decomposition to obtain $\mathbf{K}$, $\mathbf{R}$.

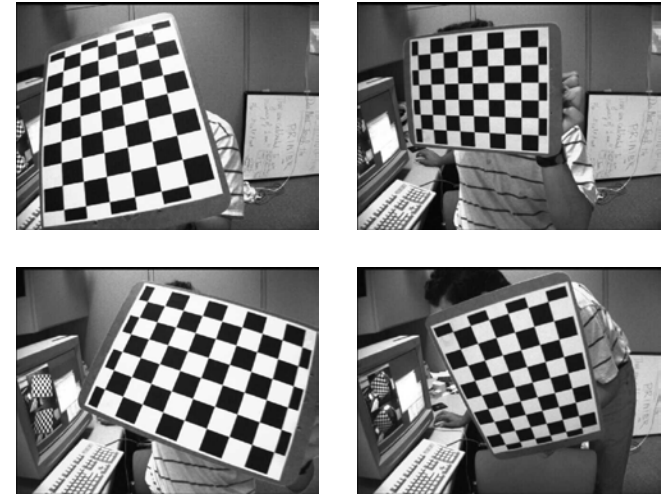
Geometric Camera Calibration

- Pros
 - Simple to formulate
 - Analytical solutions exist
- Cons
 - Does not model lens distortion
 - Hard to impose constraints
- Remarks
 - Nonlinear methods for camera calibration are preferred in practice.

$$\begin{array}{c} \text{Given} \\ \left[\begin{array}{ccc} \mathbf{X}_1^T & \mathbf{0} & -x_1 \mathbf{X}_1^T \\ \mathbf{0} & \mathbf{X}_1^T & -y_1 \mathbf{X}_1^T \\ \vdots & \vdots & \vdots \\ \mathbf{X}_n^T & \mathbf{0} & -x_n \mathbf{X}_n^T \\ \mathbf{0} & \mathbf{X}_n^T & -y_n \mathbf{X}_n^T \end{array} \right] \end{array} \begin{array}{c} \text{Unknown} \\ \left[\begin{array}{c} \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{array} \right] = \mathbf{0} \end{array}$$

Multi-Plane Calibration

- Pros
 - Only requires a plane (chessboard)
 - Can model lens distortion
- Cons
 - Need to solve nonlinear optimization problems
- Remarks
 - Multi-plane calibration considers multiple images, and minimizes the sum of the re-projection error of each image.
 - Both OpenCV and MATLAB provide software for multi-plane calibration.



Nonlinear Optimization for Multi-Plane Calibration

- Given $\mathbf{m}_{i,j} = [x_{i,j}, y_{i,j}]^T$, $\mathbf{M}_i = [X_i, Y_i, Z_i]^T$ for $i = 1, 2, \dots, n$
- $\mathbf{m}_{i,j}$ denotes the i th detected 2D points in the j th image.
- Find $\mathbf{K}$, $(\mathbf{R}_j, \mathbf{t}_j)$, (k_1, k_2) such that
 - $\tilde{\mathbf{m}}_{i,j} = L(\mathbf{K}[\mathbf{R}_j \ \mathbf{t}_j] \tilde{\mathbf{M}}_i)$ with $\tilde{\mathbf{M}}_i = \begin{bmatrix} \mathbf{M}_i \\ 1 \end{bmatrix}$, $\tilde{\mathbf{m}}_{i,j} = \lambda_{i,j} \begin{bmatrix} \mathbf{m}_{i,j} \\ 1 \end{bmatrix}$ for $i = 1, 2, \dots, n$
- Suppose $L(\mathbf{K}[\mathbf{R}_j \ \mathbf{t}_j] \tilde{\mathbf{M}}_i) = w_{i,j} [u_{i,j}, v_{i,j}, 1]^T$, and denote $\mathbf{r}_{i,j} = [u_{i,j}, v_{i,j}]^T$.
- Find $\mathbf{K}$, $(\mathbf{R}_j, \mathbf{t}_j)$, (k_1, k_2) that minimize the re-projection error:

$$\sum_{j=1}^m \sum_{i=1}^n \|\mathbf{m}_{i,j} - \mathbf{r}_{i,j}(\mathbf{K}, \mathbf{R}_j, \mathbf{t}_j, k_1, k_2, \mathbf{M}_i)\|_2^2$$

calibrateCamera() in OpenCV

```
ret, K, disc, rvecs, tvecs = calibrateCamera (3d_pts, 2d_pts, img_size, K0, disc0)
```

- 2d_pts is a list of 2D points in each image.
- K is the intrinsic matrix.
- disc is the distortion coefficients.
- rvecs is a list of rotation vectors.
- tvecs is a list of translation vectors.
- K₀ is the initial guess of K, which can be set to None.
- disc₀ is the initial guess of disc, which can be set to None.